



Simple Moving Average (SMA) Crossover Strategy



BULLISH? BEARISH

Goal: Identify buy/sell signals based on trend direction

- Bullish Signal (Buy): 50-day SMA > 200-day SMA
- Bearish Signal (Sell): 50-day SMA < 200-day SMA

50/200

How it works

- We track the closing price of a stock over time.
- We calculate two trends:
 - SMA 50: Average closing price over the last 50 days
 - SMA 200: Average closing price over the last 200 days
- When these lines cross, it signals a shift in market sentiment.



Why Sma Work

Why it works

- It smooths out price data to reveal trends
- Helps avoid noise in volatile markets
- Especially useful for long-term trend followers

Trend signal

Our buy/sell logic is simple but effective:

- Golden Cross: occurs when SMA50 crosses above SMA200
- → This is seen as a bullish reversal and suggests momentum is building upward.
- • Death Cross: happens when SMA50 crosses below SMA200
- → This indicates the short-term trend is weakening below the long-term average — a potential bearish signal.

```
import pandas as pd
import yfinance as yf
import matplotlib.pyplot as plt
import numpy as np

# Download data
ticker = "NVDA"
data = yf.download(ticker, start="2020-01-01", end="2025-01-01")
data = data[['Close']]

# Calculate SMAs
data['SMA_50'] = data['Close'].rolling(window=50).mean()
data['SMA_200'] = data['Close'].rolling(window=200).mean()

# Generate signals
data['Signal'] = np.where(data['SMA_50'] > data['SMA_200'], 1, 0)
data['Position'] = data['Signal'].diff()

# Plotting
plt.figure(figsize=(14,7))
plt.plot(data['Close'], label='NVDA Close Price', alpha=0.5)
plt.plot(data['SMA_50'], label='50-day SMA', alpha=0.9)
plt.plot(data['SMA_200'], label='200-day SMA', alpha=0.9)

# Buy signals
plt.plot(data[data['Position'] == 1].index,
         data['Close'][data['Position'] == 1],
         '^', markersize=12, color='g', label='Buy Signal')

# Sell signals
plt.plot(data[data['Position'] == -1].index,
         data['Close'][data['Position'] == -1],
         'v', markersize=12, color='r', label='Sell Signal')

plt.title('NVIDIA (NVDA) - SMA Crossover Strategy')
plt.xlabel('Date')
plt.ylabel('Price (USD)')
plt.legend()
plt.grid()
```